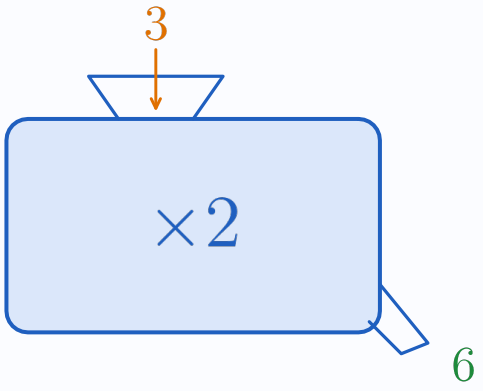
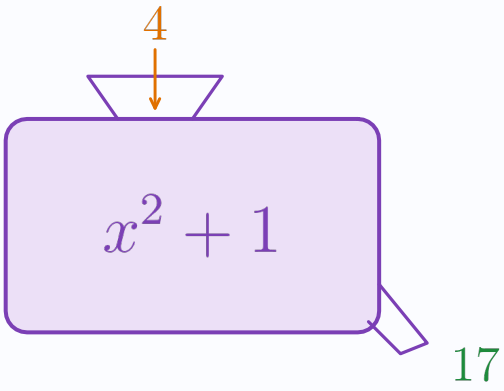
Chapter 18

Functions

a vending machine for numbers



put 3 in, 6 comes out. Every time.



put 4 in: $4^2 + 1 = 17$

A **function** is a rule that turns each input number into exactly **one** output number, like a vending machine: press B3 and you get crisps, every single time. You already know lots: "double it", "square it", "add VAT", and the sin button on your calculator.

This chapter is mostly about **reading the notation**, $f(x)$, which trips everyone up, and about which inputs a machine will accept. Then you'll chain machines together and run them backwards, which is Chapter 14's undo idea again.

Goes with **Lecture 13** (functions, domains, images, piecewise functions).

Three ways to write the same machine

RULE: function and function notation

A **function** is a rule that gives **exactly one output** for each allowed input.

$f(x)$ means "the output of f when the input is x". $f(3)$ means: replace **every** x in the rule by 3 and work it out.

$f(x)$ is **not** $f \times x$. f is the name of the rule; the brackets hold the input.

Lecture 13 writes the doubling machine three ways. They all mean "input a number, output twice it":

$x \mapsto 2x$	$y = 2x$	$f(x) = 2x$
"x maps to 2x"	"output y is 2 times x"	"f of x is 2x"

The third one, **function notation**, is the one to master. The letter f is the **name of the machine** (you could call it g, h or anything). The brackets hold **what you put in**. So:

$$f(3) = 6$$

the input → 3; machine's name → f; the output → 6

Said: "f of 3 is 6".
Meaning: put 3 into machine f, and 6 comes out.

The big trap: $f(x)$ is NOT "f times x". There is no multiplication here. f isn't a number at all, it's a name. Writing $f(3) = 3f$ is like saying "vending-machine(B3)" equals "3 vending machines".

The x is a slot: whatever goes in fills every slot

RULE: substituting into a function

Replace **every** x by the input, **in brackets**: $f(-2)$ for $f(x) = x^2 + 1$ is $(-2)^2 + 1$.

An expression can be the input: $f(a+1)$ means replace each x by $(a+1)$, then simplify.

$f(x+1)$ changes the **input**; $f(x)+1$ changes the **output**. They are different.

Take $f(x) = x^2 + 1$. Think of every x in the rule as an empty slot, $\square^2 + 1$. The input fills **every** slot, and it goes in **with brackets** round it:

$$f(\square) = \square^2 + 1$$ ← the rule with its slots showing

$$f(3) = (3)^2 + 1 = 10$$ ← 3 in the slot

$$f(-2) = (-2)^2 + 1 = 5$$ ← the brackets matter: $(-2)^2 = +4$

$$f(a+1) = (a+1)^2 + 1 = a^2 + 2a + 2$$

a whole expression can go in the slot

Watch out: $f(x+1)$ and $f(x)+1$ are different. The first puts $x+1$ **into** the machine; the second adds 1 to what comes **out**. With $x = 3$: $f(4) = 17$, but $f(3)+1 = 11$.

If you code: a function in maths is exactly a Python function. The letter in the brackets is the parameter; calling it with a number fills the slot.

```
def f(x):
    return x**2 + 1

print(f(3))    # 10
print(f(-2))   # 5
```

def = define the machine

f(3) = use it with input 3

The domain: which coins the machine accepts

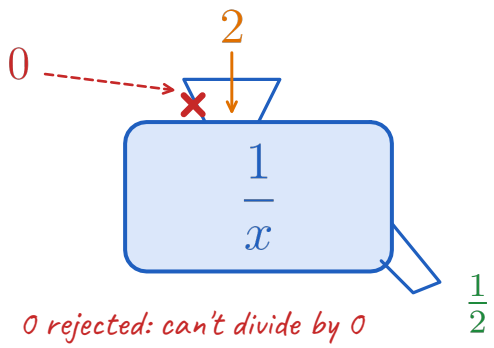
RULE: domain and range

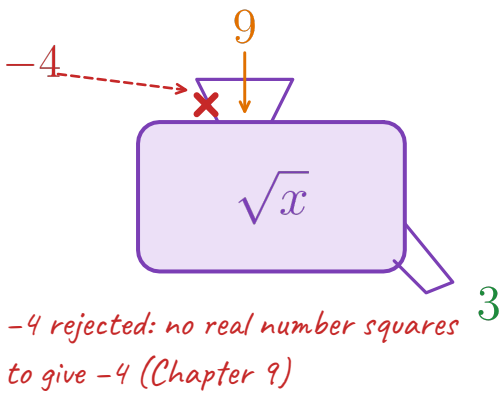
Domain = the set of inputs allowed. **Range** (or image) = the set of outputs that actually come out.

If no domain is given, take every real number **except** those that break the rule: a denominator can't be 0, and you can't square-root a negative number.

A function can be given a **restricted domain** on purpose, e.g. $f(x) = x^2,\ x \geq 0$. The domain is part of the function: change it and you change the range.

A vending machine rejects foreign coins. Some functions reject some inputs, because the rule would break:





The set of inputs a function accepts is its **domain**. The set of outputs that can come out is its **image** (also called its **range**). Lecture 13's examples, in the set notation of Chapter 6:

function	domain	image	why
$f(x) = 2x$	$\mathbb{R}$	$\mathbb{R}$	anything in, anything out
$f(x) = x^2$	$\mathbb{R}$	$[0, \infty)$	squares are never negative
$f(x) = \sin x$	$\mathbb{R}$	$[-1, 1]$	the sin button never gives more than 1
$f(x) = \frac{1}{x}$	$x \neq 0$	$y \neq 0$	no input makes 1/x equal 0

(Lecture 13 writes the squaring function's image as $\mathbb{Q}^+$, but that's the positive **fractions**; the outputs are all the non-negative **real** numbers, $[0, \infty)$, including 0 and numbers like $(\sqrt[3]{2})^2$.)

When no domain is given, assume it's **every number that doesn't break the rule**: usually just avoid dividing by zero and square-rooting negatives.

PROCEDURE: finding the domain and the range

1. **Domain:** set any denominator $\neq 0$ and anything under a square root ≥ 0; solve for x.
2. **Range of a straight line on an interval:** work out the outputs at the two ends of the domain. The range runs between them.
3. **Range of a square:** $(\text{anything})^2 \geq 0$, so $a(x-h)^2 + k \geq k$ when $a > 0$ (Chapter 15's completed square). The smallest output is k.
4. Check with one or two inputs that the outputs really land in your range.

$$g(x) = \sqrt{x-3}: \quad x - 3 \geq 0 \;\Rightarrow\; x \geq 3$$ ← domain: no negatives under the root

$$h(x) = \frac{5}{2x-6}: \quad 2x - 6 \neq 0 \;\Rightarrow\; x \neq 3$$ ← domain: denominator not 0

$$f(x) = 2x + 1,\ 0 \leq x \leq 3: \quad f(0) = 1,\ f(3) = 7$$

range: work out the two ends

$$\Rightarrow\; 1 \leq f(x) \leq 7$$ ← the outputs run between them

$$q(x) = x^2 - 6x + 11 = (x-3)^2 + 2 \geq 2$$ ← range: f(x) ≥ 2 (complete the square)

Is it a function? One button, one snack

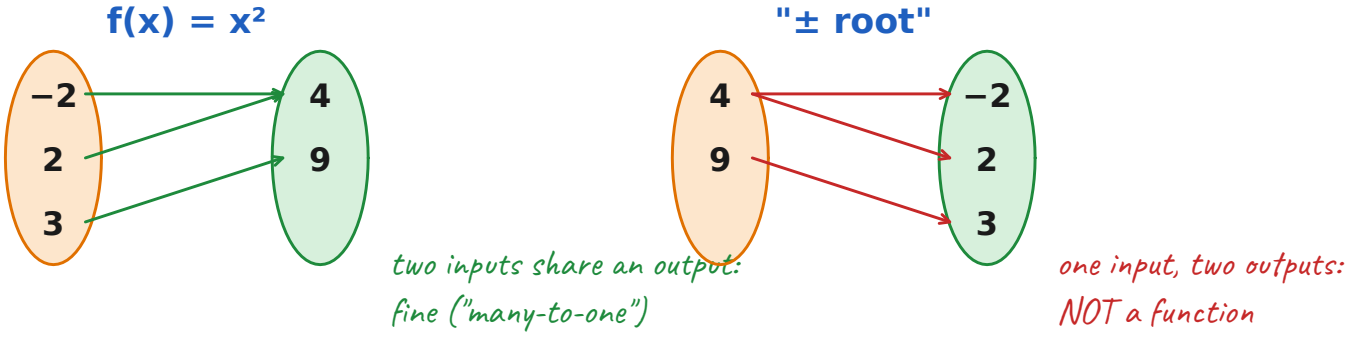
RULE: function tests

A rule is a function only if every input in the domain gives **exactly one** output.

Vertical line test (on a graph, Chapter 19): every vertical line meets the graph **at most once**. Crossing twice means one input has two outputs: not a function.

One-to-one: different inputs always give different outputs (every horizontal line meets the graph at most once). **Many-to-one**: some output comes from two or more inputs, like x^2.

A machine that gave crisps **or** chocolate when you pressed B3 would be broken. A function must give **exactly one output per input**. Draw arrows from inputs to outputs:

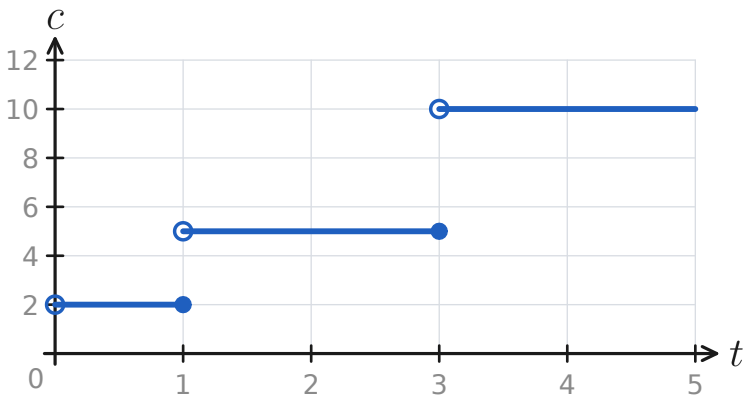


That's why $\sqrt{\ }$ on its own always means the **positive** root (Chapter 9): it keeps the square-root button a proper function.

Piecewise functions: parking charges

Some rules change depending on the input. A car park charges £2 for up to 1 hour, £5 for up to 3 hours, and £10 for longer. One machine, three rules:

$$c(t) = \begin{cases} 2 & 0 < t \le 1 \\ 5 & 1 < t \le 3 \\ 10 & t > 3 \end{cases}$$ ← *t = hours parked, c = cost in £*



filled dot = included
hollow dot = not included

exactly 1 hour costs £2 (filled dot on the lower step); 1 hour 1 minute costs £5

To use it, **first find which piece your input belongs to**, then use that piece's rule: $c(0.5) = 2$, $c(3) = 5$, $c(4.5) = 10$.

Lecture 13's example is the **sign function**: 1 for positive inputs, −1 for negative inputs, 0 for 0. It can't be written as one formula, so it's written in pieces:

$$\operatorname{sgn}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$ ← *sgn(7) = 1, sgn(−0.2) = −1*

Machines in a chain: composite functions

RULE: composite functions

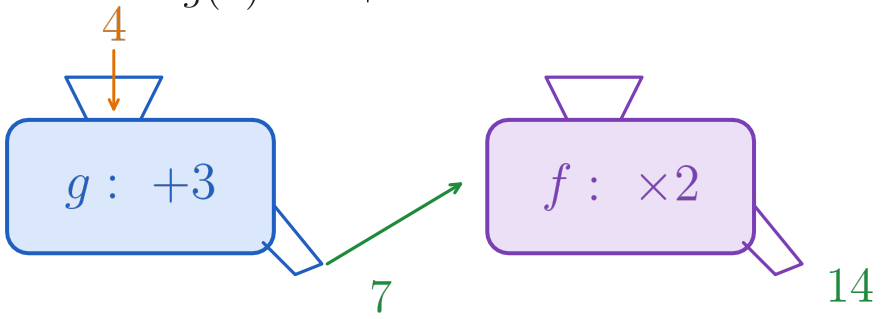
$fg(x)$ means $f(g(x))$: **do g first**, then put its output into f. It is also written $(f \circ g)(x)$, read "f after g".

Work from the **inside out**: the function written nearest to x acts first.

In general $fg(x) \neq gf(x)$: the order matters.

To find $fg(x)$ as a formula: take the rule for f and replace every x by the whole of $g(x)$, in brackets.

The output of one machine can be fed straight into another. With $f(x) = 2x$ and $g(x) = x + 3$:



g first, then f. Written f(g(4)): read from the inside out

$$fg(x) = f(g(x)) = 2(x+3) = 2x + 6$$ ← *g's output fills f's slot*

$$gf(x) = g(f(x)) = 2x + 3$$ ← *other order: different machine!*

Order matters, like socks and shoes (Chapter 14). $f(g(4)) = 14$ but $g(f(4)) = 11$.

With squares: $f(x) = x^2 + 1$ and $g(x) = 3x - 2$.

$$fg(x) = (3x-2)^2 + 1 = 9x^2 - 12x + 5$$ ← *replace x in f by (3x − 2)*

$$gf(x) = 3(x^2+1) - 2 = 3x^2 + 1$$ ← *replace x in g by $(x^2 + 1)$*

Check with a number: $x = 1$: $g(1) = 1$, $f(1) = 2$; formula: $9 - 12 + 5 = 2$ ✓.

Running a machine backwards: inverse functions

RULE: inverse functions

f^{-1} undoes f: if $f(a) = b$ then $f^{-1}(b) = a$. So $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$.

Only a **one-to-one** function has an inverse. A many-to-one function (like x^2 on all numbers) needs its domain restricted first (e.g. $x \geq 0$).

Domain of f^{-1} = range of f, and range of f^{-1} = domain of f: inputs and outputs swap.

The graph of $y = f^{-1}(x)$ is the graph of $y = f(x)$ **reflected in the line $y = x$**.

$f^{-1}(x)$ is **not** $\dfrac{1}{f(x)}$: the -1 is a label meaning "reverse".

PROCEDURE: finding f⁻¹(x)

1. Write $y = f(x)$.
2. Rearrange to make x the subject (undo each step in reverse order, Chapter 14).
3. Swap the letters: write the answer with x as the input, $f^{-1}(x) = \ldots$
4. State the domain of f^{-1} (the range of f) and check: $f^{-1}(f(a)) = a$ for some number a.

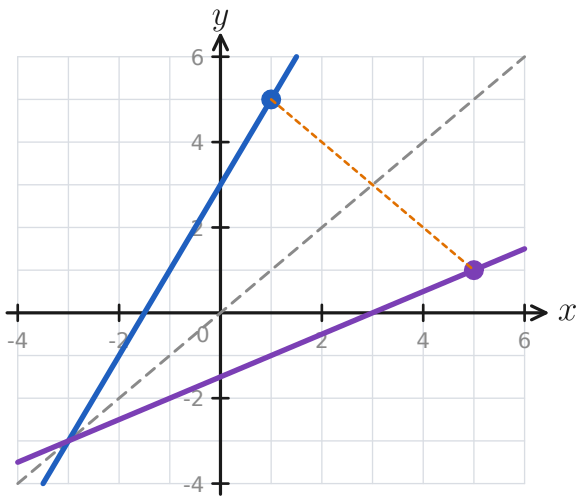
Example: $f(x) = 2x + 3$ (× 2, then + 3).

$$y = 2x + 3$$ ← step 1

$$x = \frac{y - 3}{2}$$ ← step 2: undo +3, then undo ×2

$$f^{-1}(x) = \frac{x - 3}{2}$$ ← step 3: rename the input x

Step 4, check: $f(5) = 13$ and $f^{-1}(13) = \frac{10}{2} = 5$ ✓. Domain of f^{-1}: every number, since f's range is every number.



blue: $y = 2x + 3$
purple: $y = (x - 3)/2$
grey dashed: $y = x$ (the mirror)

(1, 5) on f becomes (5, 1) on the inverse: the coordinates swap, which is a reflection in $y = x$.

Example: $f(x) = \dfrac{x+4}{3}$ with domain $x \geq 2$.

$$y = \frac{x+4}{3} \;\Rightarrow\; 3y = x + 4 \;\Rightarrow\; x = 3y - 4$$

× 3, then − 4

$$f^{-1}(x) = 3x - 4, \quad x \geq 2$$ ← domain of the inverse = range of f

The range of f: the smallest input is 2, giving $f(2) = \frac{6}{3} = 2$, and f goes up, so $f(x) \geq 2$. That becomes the domain of f^{-1}.

Example: $f(x) = x^2$ has no inverse on all numbers: $f(3) = f(-3) = 9$, so $f^{-1}(9)$ couldn't decide. Restrict to $x \geq 0$ and it is one-to-one, with $f^{-1}(x) = \sqrt{x}$, $x \geq 0$.

Find the unknown: working backwards from outputs

PROCEDURE: partial-information function problems

1. "Solve $f(x) = k$": write the rule $= k$ and solve the equation for x (Chapter 15). A square may give two answers; check each is in the domain.
2. "$f(1) = 5$" is an equation: put $x = 1$ into the rule and set it equal to 5. Two unknown letters need two facts: solve them simultaneously.
3. "Solve $fg(x) = k$": first write $fg(x)$ as one formula, then set it equal to k.
4. "Find $f^{-1}(b)$": either find the formula for f^{-1}, or solve $f(x) = b$; it's the same answer.

Example: $f(x) = 3x - 5$. Solve $f(x) = 10$.

$$3x - 5 = 10 \Rightarrow 3x = 15 \Rightarrow x = 5$$ ← check: f(5) = 15 − 5 = 10

Example: $f(x) = x^2 - 2x$. Solve $f(x) = 3$.

$$x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$$

rearrange to = 0, factorise

$$x = 3 \text{ or } x = -1$$ ← two inputs give 3: f is many-to-one

Example: $f(x) = ax + b$, with $f(1) = 5$ and $f(3) = 11$. Find a and b.

$$a + b = 5, \qquad 3a + b = 11$$ ← x = 1 and x = 3 into the rule

$$2a = 6 \Rightarrow a = 3, \quad b = 2$$ ← subtract: b cancels

Check: $f(x) = 3x + 2$: $f(1) = 5$ ✓, $f(3) = 11$ ✓.

Example: $f(x) = 2x + 1$ and $g(x) = x^2$. Solve $fg(x) = 19$.

$$fg(x) = f(x^2) = 2x^2 + 1$$ ← g first, then f

$$2x^2 + 1 = 19 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$$

Example: $f(x) = kx^2 - 3$ and $f(2) = 9$. Find k.

$$4k - 3 = 9 \;\Rightarrow\; k = 3$$ ← *x = 2 into the rule*

Example: $f(x) = 4x - 7$. Find $f^{-1}(5)$ without finding the formula.

$$4x - 7 = 5 \;\Rightarrow\; x = 3, \text{ so } f^{-1}(5) = 3$$ ← *the input that gives 5*

Mistakes to avoid

✗ Common mistakes

✗ $f(3) = 3f$

f is a name, not a number

✓ $f(3) =$ rule with $x = 3$

✗ $f(-2) = -2^2 + 1 = -3$

brackets round the input

✓ $f(-2) = (-2)^2 + 1 = 5$

✗ $fg(x)$: do f first

inside out

✓ $fg(x) = f(g(x))$: g first

✗ $f^{-1}(x) = \dfrac{1}{f(x)}$

−1 means inverse, not a power

✓ f^{-1} undoes f

✗ $f(x+1) = f(x) + 1$

input changed vs output changed

✓ $f(x+1) \neq f(x) + 1$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Function	exactly one output for each input in the domain
Notation	$f(a) =$ the rule with every x replaced by (a)
Domain	allowed inputs: denominator $\neq 0$, nothing negative under a square root
Range	all outputs; for a line on an interval use the end values; $a(x-h)^2 + k \geq k$ $(a > 0)$
Vertical line test	a graph is a function if every vertical line meets it at most once
One-to-one	different inputs give different outputs; needed for an inverse
Composite	$fg(x) = (f \circ g)(x) = f(g(x))$ g first; in general $fg \neq gf$
Inverse	$f(f^{-1}(x)) = f^{-1}(f(x)) = x$ write $y = f(x)$, make x the subject, rename
Inverse domain	domain of f^{-1} = range of f (and range of f^{-1} = domain of f)
Inverse graph	reflect the graph of f in the line $y = x$
Solve f(x) = k	set the rule equal to k and solve; $f(a) = b$ gives an equation

Vending machines: your turn

Answers in grey.

Using the notation

1. $f(x) = 3x - 1$. Find $f(2)$, $f(0)$ and $f(-4)$.
2. $g(x) = x^2 - x$. Find $g(3)$ and $g(-3)$.
3. $h(x) = 10/x$. Find $h(5)$ and $h(0.5)$.
4. $f(x) = 3x - 1$. Find $f(a + 2)$ and simplify.

1. 5, −1, −13 2. 6, 12 3. 2, 20 4. 3a + 5

Domain and image

5. What input does $1/(x - 4)$ reject?
6. What inputs does $\sqrt{(x - 1)}$ accept?
7. What is the image of $f(x) = x^2 + 5$?

5. $x = 4$ 6. $x \geq 1$ 7. $[5, \infty)$: outputs of 5 or more

Pieces, chains and inverses

8. With the car park $c(t)$, find $c(2)$ and $c(3.5)$.
9. $f(x) = x + 4$, $g(x) = 3x$. Find $f(g(2))$ and $g(f(2))$.
10. Find the inverse of $f(x) = 5x - 2$.
11. Find the inverse of $f(x) = (x + 1)/4$.

8. £5, £10 9. 10, 18 10. $f^{-1}(x) = (x + 2)/5$ 11. $f^{-1}(x) = 4x - 1$

Find the unknown

12. $f(x) = 5x + 2$. Solve $f(x) = 27$.
13. $g(x) = x^2 + 4x$. Solve $g(x) = 12$.
14. $f(x) = ax + b$, with $f(2) = 7$ and $f(5) = 16$. Find a and b.
15. $f(x) = 3x - 1$ and $g(x) = x + 2$. Solve $fg(x) = 20$.
16. $h(x) = kx^3 + 1$ and $h(2) = 25$. Find k.
17. $f(x) = (x + 4)/2$. Find the value of a for which $f^{-1}(a) = 6$.

12. $x = 5$ ($5x = 25$) 13. $x = 2$ or $x = -6$ ($(x + 6)(x - 2) = 0$) 14. $a = 3$, $b = 1$ ($3a = 9$) 15. $x = 5$ ($fg(x) = 3(x + 2) - 1 = 3x + 5$) 16. $k = 3$ ($8k + 1 = 25$) 17. $a = 5$ ($f^{-1}(a) = 6$ means $a = f(6)$)